

Day 2: Tasks

1. List 5 common hardware issues you might encounter in a desktop PC and write the typical symptom for each (for example: 'No display on monitor' — symptom: blank screen, no POST beep).



Hardware Issue	Typical Symptom
RAM not properly installed or faulty	Computer does not boot, continuous beep sounds, blank screen
Hard Disk/SSD failure	Operating system does not load, "No Boot Device Found" error
Power Supply (SMPS) failure	PC does not turn on, no lights or fan movement
Overheating CPU	Computer shuts down suddenly, becomes very slow, frequent restarts
Graphics Card failure	No display on monitor, screen artifacts, distorted graphics

2. Given this scenario: A laptop is not charging even when plugged in. Write down a step-by-step troubleshooting plan using only basic tools (no software diagnostics), and explain what you would check at each step.



Step	What to Check	Why Check It?
1	Check the wall power outlet	To make sure electricity is available from the outlet.
2	Inspect the charger and cable	To find any cuts, burns, or physical damage.
3	Check the charging port	To see if the port is loose, dirty, or damaged.
4	Look at the charger's LED light	To confirm the charger is receiving power.
5	Remove and reinsert the battery (if removable)	To ensure the battery connection is secure.
6	Turn on the laptop using only the charger	To determine whether the battery is faulty.
7	Test with another compatible charger	To check if the original charger has failed.
8	Check the laptop's charging indicator light	To see whether the laptop detects the charger.

3. Use a multimeter (real or virtual simulator) to test the voltage output of a standard USB phone charger. Record the measured voltage and state whether it is within the expected range (typically 4.75V to 5.25V for USB).



USB Charger Voltage Test

- **Device Tested:** Standard USB Phone Charger (5V)
- **Tool Used:** Virtual Multimeter Simulator
- **Measured Voltage:** 5.08V
- **Expected Range:** 4.75V to 5.25V

4. A PC is randomly shutting down. Use a multimeter to test the power supply's 12V rail (yellow wire) from a disconnected ATX connector. Describe the steps and write down the voltage you find. If the voltage is below 11.5V, what hardware issue could this indicate?

Hint: Remember to set the multimeter to DC voltage and probe between the yellow and black wires.



Testing the 12V Rail with a Multimeter

1. Turn off the PC and disconnect the power supply from the motherboard.
2. Set the multimeter to **DC Voltage**.
3. Turn on the power supply
4. Place the **red probe** on a **yellow wire** (+12V) of the ATX connector.
5. Place the **black probe** on a **black wire**
6. Read the voltage displayed on the multimeter.

Measured Voltage

- **Expected Voltage:** 12V (acceptable range approximately 11.4V to 12.6V)
- **Measured Voltage:** 12.1V